

Carrier-frequency estimation of the *DHDDS* c.124A>G (p.Lys42Glu) founder allele in 7,738 self-identified Ashkenazi Jewish adults: results from the Hadassah Reproductive Genetics Screening Program, 2010–2014

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ABSTRACT

Background. The *DHDDS* c.124A>G (p.Lys42Glu) variant is a recognized Ashkenazi Jewish (AJ) founder allele causing autosomal-recessive retinitis pigmentosa type 59 (RP59). Reliable carrier-frequency data in the AJ population are needed to inform reproductive-genetics counselling. **Methods.** Between 2010 and 2014, the Hadassah Reproductive Genetics Screening Program offered targeted testing for the c.124A>G allele to consecutive self-identified AJ adults presenting for premarital or preconception counselling. Testing used a TaqMan allelic-discrimination assay (Applied Biosystems, assay C__34289572_10), validated against Sanger sequencing in 200 randomly selected samples (100% concordance). **Results.** Of **7,738** consecutively screened individuals (4,089 women, 3,649 men; mean age 27.3 years), **24 carrier alleles** were identified, all in heterozygous state. No homozygote was identified in the screening population. The observed carrier frequency was **1 in 322** individuals ($24/7,738 = 0.310\%$), with a corresponding population allele frequency of 1.55×10^{-3} . Stratification by sub-population (Polish AJ vs Lithuanian AJ vs Hungarian AJ) did not reveal statistically significant heterogeneity ($\chi^2 = 1.84$, $p = 0.40$). **Conclusions.** The c.124A>G carrier frequency in the contemporary AJ population is consistent with prior published estimates and supports inclusion of the variant in expanded AJ carrier-screening panels.

KEY NUMBERS (BOX 1)

Quantity	Value
Total individuals screened	7,738
Carrier (heterozygous) alleles detected	24
Homozygous individuals identified	0
Observed carrier frequency	1 in 322 (0.310%)
Estimated population allele frequency	1.55×10^{-3}
Number of probands referred onward to ophthalmology	0 (asymptomatic carriers only)
Time span of screening	January 2010 – December 2014

Note: The carrier alleles enumerated in this screening study are **asymptomatic carrier alleles in an unrelated reproductive-genetics screening population**. They are **NOT** the same as proband-borne pathogenic alleles in any clinical IRD cohort and should not be conflated with cohort-specific allele tallies reported elsewhere in the literature.

DISCUSSION (excerpt)

Our observed carrier frequency of 1 in 322 closely matches the original estimate by Züchner et al. (2011) of 1 in 322 in the United States AJ population, and the gnomAD v2.0.2 AJ subset estimate of 1 in 317. This convergence across three independent populations argues that *DHDDS* c.124A>G is a stable AJ founder allele and that population-level prevalence estimates derived from any of these screens are mutually consistent.

Of note, we identified **zero homozygous individuals** in the 7,738 screened adults. Given an allele frequency of 1.55×10^{-3} , the expected number of homozygotes under Hardy–Weinberg equilibrium in a sample of this size is approximately $7,738 \times (1.55 \times 10^{-3})^2 \approx 0.019$, so zero observed homozygotes is fully consistent with the predicted expectation.

Clinical implication. Couples in which both partners are carriers face a 1-in-4 risk per pregnancy of an affected child; reproductive options including pre-implantation genetic diagnosis are routinely offered through our service.